2.52 Forces: Equilibrium

Worked Solutions

1 a)

x-comp: $5\cos 30° = 4.33$ N (3 SF)

y-comp: $5\sin 30° = 2.5$ N

1 b)

x-comp: $6\cos 45° = 4.24$ N (3 SF)

y-comp: $6\sin 45° = 4.24$ N (3 SF)

1 c)

x-comp: $8\cos 40° = 6.13$ N (3 SF)

y-comp: $8\sin 40° = 5.14$ N (3 SF)

1 d)

x-comp: $3\cos 22° = 2.78$ N (3 SF)

y-comp: $3\sin 22° = 1.12$ N (3 SF)

2 a)

x-comp: $10\sin 45° = 7.07$ N (3 SF)

y-comp: $10\cos 45° = 7.07$ N (3 SF)

2 b)

x-comp: $-12\cos 20° = -11.3$ N (3 SF)

y-comp: $12\sin 20° = 4.10$ N (3 SF)

2 c)

x-comp: $5\cos 30° = 4.33$ N (3 SF)

y-comp: $-5\sin 30° = -2.5$ N

2 d)

x-comp: $-14\cos 35° = -11.5$ N (3 SF)

y-comp: $-14\sin 35° = -8.03$ N (3 SF)

2 e)

x-comp: $25\cos 16° = 24.0$ N (3 SF)

y-comp: $25\sin 16° = 6.89$ N (3 SF)

2 f)

x-comp: $3\sin 70° = 2.82$ N (3 SF)

y-comp: $-3\cos 70° = -1.03$ N (3 SF)

2 g)

x-comp: $-10\sin 64° = -8.99$ N (3 SF)

y-comp: $-10\cos 64° = -4.38$ N (3 SF)

2 h)

x-comp: $-7\sin 28° = -3.29$ N (3 SF)

y-comp: $7\cos 28° = 6.18$ N (3 SF)

3 a)

x-comp: $10 - 4\cos 34° = 6.68384971\ldots$

y-comp: $2 + 4\sin 34° = 4.236771614\ldots$

Magnitude: $\sqrt{44.673\ldots + 17.950\ldots} = 7.91$ N (3 SF)

Angle with Ox: $\tan^{-1}\left(\frac{4.236771614\ldots}{6.68384971\ldots}\right) = 32.4°$ (3 SF)

3 b)

x-comp: $8\cos 60° - 6 = -2$

y-comp: $8\sin 60° - 3 = -3 + 4\sqrt{3} = 3.9282\ldots$

Magnitude: $\sqrt{4 + 15.430\ldots} = 4.41$ N (3 SF)

Angle with Ox: $180° - \tan^{-1}\left(\frac{3.9282\ldots}{2}\right) = 117°$ (3 SF)

3 c)

x-comp: $7\cos 45° - 7\cos 45° = 0$

y-comp: $7\sin 45° + 7\sin 45° - 5$

$= -5 + 7\sqrt{2} = 4.8999494937\ldots$

Magnitude: 4.90 N (3 SF)

Angle with Ox: $90°$

4 a)

$P - 10\sin 60° = 0 \Rightarrow P = 5\sqrt{3} = 8.66$ (3 SF)

$Q - 10\cos 60° = 0 \Rightarrow Q = 5$

4 b)

$P - 6\sin 30° = 0 \Rightarrow P = 3$ N

$6\cos 30° - Q = 0 \Rightarrow Q = 3\sqrt{3} = 5.20$ (3 SF)

4 c)

$P\sin 60° - 12\sin 75° = 0$

$\Rightarrow P = 2\sqrt{6} + 6\sqrt{2} = 13.4$ (3 SF)

$P\cos 60° + 12\cos 75° - Q = 0$

$\Rightarrow Q = 4\sqrt{6} = 9.80$ (3 SF)

4 d)

$-P\cos 40° + Q\cos 60° = 0$

$P\sin 40° + Q\sin 60° - 30 = 0$

$\Rightarrow P = 15.2$ (3 SF) and $Q = 23.3$ (3 SF)

4 e)

$-P\cos 23° + Q\sin 50° = 0$

$P\sin 23° + Q\cos 50° - 32 = 0$

$\Rightarrow P = 27.5$ (3 SF) and $Q = 33.1$ (3 SF)

4 f)

$6\cos\theta° - 4 = 0 \Rightarrow \theta = 48.2°$ (3 SF)

$P - 6\sin\theta° = 0 \Rightarrow P = 4.47$ (3 SF)

4 g)

$15\cos 30° - 22\sin\theta° = 0 \Rightarrow \theta = 36.2$ (3 SF)

$P - 15\sin 30° - 22\cos\theta° = 0 \Rightarrow P = 25.3$ (3 SF)

4 h)

$P\cos\theta° - 20\sin 65° = 0 \Rightarrow P\cos\theta° = 20\sin 65°$

$P\sin\theta° + 20\cos 65° - 40 = 0$

$\Rightarrow P\sin\theta° = 40 - 20\cos 65°$

$\Rightarrow \theta = 60.1$ (3 SF), $P = 36.4$ (3 SF)

4 i)

$P\cos\theta° + 4\cos 42° - 8 = 0$

$\Rightarrow P\cos\theta° = 8 - 4\cos 42°$

$-P\sin\theta° + 4\sin 42° = 0 \Rightarrow P\sin\theta° = 4\sin 42°$

$\Rightarrow \theta = 28.0$ (3 SF), $P = 5.70$ (3 SF)

4 j)

$P - 100\cos 45° = 0$

$\Rightarrow P = 100\cos 45° = 50\sqrt{2} = 70.7$ N (3 SF)

$Q - 100\sin 45° = 0$

$\Rightarrow Q = 100\sin 45° = 50\sqrt{2} = 70.7$ N (3 SF)

4 k)

$P - 50\cos 35° = 0$

$\Rightarrow P = 50\cos 35° = 41.0$ N (3 SF)

$Q - 50\sin 35° = 0$

$\Rightarrow Q = 50\sin 35° = 28.7$ N (3 SF)

5 a)

$T_2 \cos 45° - T_1 \cos 35° = 0$

$T_2 \sin 45° + T_1 \sin 35° - 2 \times 9.8 = 0$

$\Rightarrow T_1 = 14.1$ (3 SF) and $T_2 = 16.3$ (3 SF)

5 b)

$T_2 \sin 42° - T_1 \sin 18° = 0$

$T_2 \cos 42° + T_1 \cos 18° - 2 \times 9.8 = 0$

$\Rightarrow T_1 = 15.1$ (3 SF) and $T_2 = 6.99$ (3 SF)

5 c)

$T_2 \cos 50° - T_1 \cos 50° = 0$

$T_2 \sin 50° + T_1 \sin 50° - 2 \times 9.8 = 0$

$\Rightarrow T_1 = 12.8$ (3 SF) and $T_2 = 12.8$ (3 SF)

6 a)

$T - 14 \sin 70° = 0 \Rightarrow T = 13.2$ (3 SF)

$14 \cos 70° - 9.8m = 0 \Rightarrow m = 0.487$ (3 SF)

6 b)

$5 - T \cos 60° = 0 \Rightarrow T = 10$

$10 \sin 60° - 9.8m = 0 \Rightarrow m = 0.884$ (3 SF)

7 a)

$15 \sin \theta° - 10 = 0 \Rightarrow \theta = 41.8$ (3 SF)

7 b)

$T - 15 \cos \theta° = 0 \Rightarrow T = 11.2$ (3SF)

8 a)

$18 - T \cos \theta° = 0 \Rightarrow T \cos \theta° = 18$

$T \sin \theta° - 4 = 0 \Rightarrow T \sin \theta° = 4$

$\theta = 12.5$ (3 SF)

8 b)

$T = 18.4$ (3 SF)

9 a)

$T_1 \cos \theta° - 40 = 0 \Rightarrow T_1 = \frac{40}{\cos \theta°}$ N

9 b)

$T_2 - \frac{40}{\cos \theta°} \times \sin \theta° = 0 \Rightarrow T_2 = 40 \tan \theta°$

9 c)

$\frac{40}{\cos\theta^\circ} = 40\frac{\sin\theta^\circ}{\cos\theta^\circ} \Rightarrow \sin\theta = 1 \Rightarrow 90^\circ$

9 d)

$40\tan\theta^\circ < 40 \Rightarrow \tan\theta^\circ < 1 \Rightarrow \theta < 45^\circ$

$\therefore 0^\circ \leq \theta < 45^\circ$

10 a)

$T\sin\theta^\circ + T\sin\theta^\circ - mg = 0$

$\Rightarrow 2T\sin\theta^\circ = mg$

$\Rightarrow T = \frac{mg}{2\sin\theta^\circ}$ as req.

10 b)

$T = \frac{5}{2\sin 45^\circ} = \frac{5}{2}\sqrt{2} = 3.54$ N (3SF)

10 c)

$T = \frac{5}{2\sin 60^\circ} = \frac{5}{3}\sqrt{3} = 2.89$ N (3SF)

10 d)

As $\theta \to 90^\circ, T \to 2.5$ N

11 a)

$1.6^2 = 1.4^2 + 2^2 - 2\times 1.4\times 2\times\cos\alpha$

$\Rightarrow \cos\alpha = \frac{1.6^2-1.4^2-2^2}{-2\times 1.4\times 2} = \frac{17}{28}$ as req.

11 b)

$1.4^2 = 1.6^2 + 2^2 - 2\times 1.6\times 2\times\cos\beta$

$\Rightarrow \cos\beta = \frac{1.4^2-1.6^2-2^2}{-2\times 1.6\times 2} = \frac{23}{32}$ as req.

11 c)

$R(\rightarrow)$: $T_B\cos\beta - T_A\cos\alpha = 0$

$\frac{23}{32}T_B - \frac{17}{28}T_A = 0$

$\frac{23}{32}T_B = \frac{17}{28}T_A$

$\frac{T_B}{T_A} = \frac{\frac{17}{28}}{\frac{23}{32}} = \frac{136}{161}$ as req.

11 d)

$T_B = 272$ N

$R(\uparrow)$: $322\sin\alpha + 272\sin\beta - mg = 0$

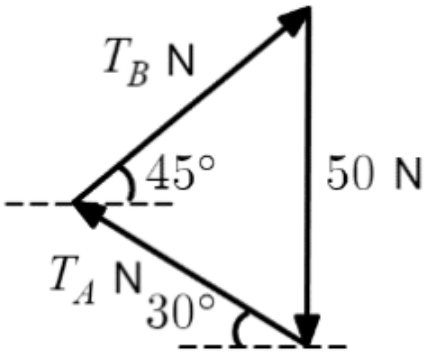
$\sqrt{28^2-17^2} = 3\sqrt{55}$ so $\sin\alpha = \frac{3\sqrt{55}}{28}$

$\sqrt{32^2-23^2} = 3\sqrt{55}$ so $\sin\beta = \frac{3\sqrt{55}}{32}$

$R(\uparrow)$: $322\times\frac{3\sqrt{55}}{28} + 272\times\frac{3\sqrt{55}}{32} - mg = 0$

$60\sqrt{55} - mg = 0 \Rightarrow m = \frac{60\sqrt{55}}{g}$ kg

12 a)

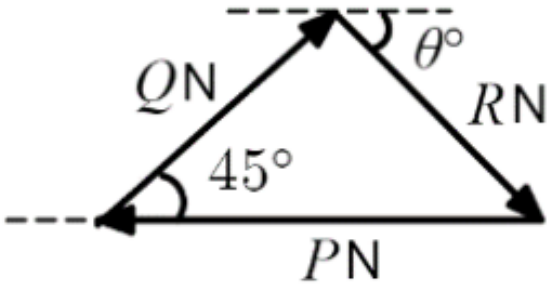


12 b)

$\frac{50}{\sin 75°} = \frac{T_B}{\sin 60°} \Rightarrow T_B = 44.8$ N

$\frac{50}{\sin 75°} = \frac{T_A}{\sin 45°} \Rightarrow T_A = 36.6$ N

13 a)

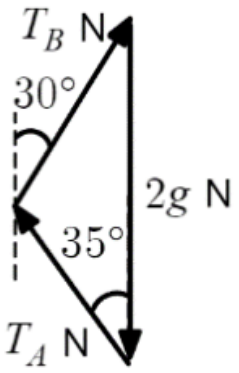


13 b)

$180° - 45° = 135°$

$0 < \theta < 135$

14 a)



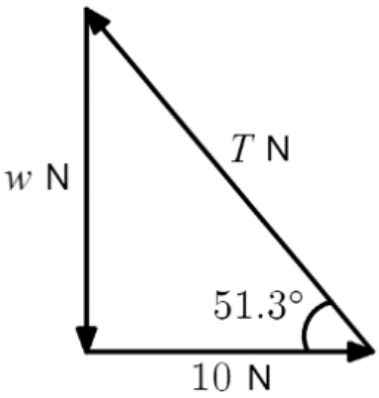
14 b)

$\frac{19.6}{\sin 115°} = \frac{T_B}{\sin 35°} \Rightarrow T_B = 12.4$ N

$\frac{19.6}{\sin 115°} = \frac{T_A}{\sin 30°} \Rightarrow T_A = 10.8$ N

15 a)

$\cos^{-1}\left(\frac{5}{8}\right) = 51.3°$



15 b)

$T = \frac{10}{\frac{5}{8}} = 16$ N

15 c)

$w = \sqrt{16^2 - 10^2} = 2\sqrt{39}$ N

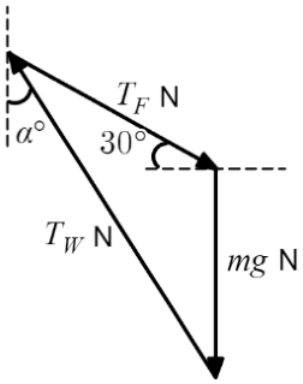
$w = 12.5$ N

15 d)

Mass is $\frac{2\sqrt{39}}{9.8} = 1.27$ kg

1.3 kg (2 SF – AQA)

16 a)



16 b)

$\frac{T_W}{\sin 120°} = \frac{mg}{\sin 30°} \Rightarrow T_W = \frac{mg \sin 120°}{\sin 30°} = \sqrt{3}mg$ as req.

16 c)

$\frac{T_F}{\sin 30°} = \frac{mg}{\sin 30°} \Rightarrow T_F = mg$ as req.

16 d)

$\frac{T_W}{\sin 135°} = \frac{mg}{\sin(45-\alpha)°}$

$\sin(45-\alpha)° = \sin 45° \cos \alpha° - \cos 45° \sin \alpha°$

$= \frac{\sqrt{2}}{2}(\cos \alpha° - \sin \alpha°)$

$\therefore \frac{T_W}{\frac{\sqrt{2}}{2}} = \frac{mg}{\frac{\sqrt{2}}{2}(\cos \alpha° - \sin \alpha°)}$

which gives $T_W = \frac{mg}{\cos \alpha° - \sin \alpha°}$ as req.